CHAPTER 10 EAMCET MATHEMATICS

PROPERTIES OF TRIANGLES, HEIGHTS AND DISTANCES

SYNOPSIS AND FORMULAE

Properties of Triangle

In a ΔABC, if a, b, c are its sides and A, B and C are its angles then,

(i) Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Where,

R = Circum radius

The ratio of sides of triangle is equal to respective ratio of sines of angles i.e.,

$a : b : c = \sin A : \sin B : \sin C$.

(ii) Cosine Rule (or) Law of Cosines

In ΔABC,

$$b^2 = c^2 + a^2 - 2ca \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

(iii) Deductions from Cosine Rule

+ $\cos A = \dfrac{b^2+c^2-a^2}{2bc}, \cos B = \dfrac{c^2+a^2+b^2}{2ac}, \cos C = \dfrac{a^2+b^2-c^2}{2ab}$
+ $a^2 + b^2 + c^2 = 2ab \cos C + 2bc \cos A + 2ca \cos B$

(iv) Projection Rule

In ΔABC,

$$a = b \cos C + c \cos B$$

$$b = c \cos A + a \cos C$$

$$c = a \cos B + b \cos A$$

(v) Deductions from Projection Rule

$2s = a + b + c$

$= (b+c) \cos A + (c+a) \cos B + (a+b) \cos C$

$2s = a + b + c$

$= a(\cos B + \cos C) + b(\cos C + \cos A) + c(\cos A + \cos B)$

(vi) Napier Analogy (or) Tangent Rule

In ΔABC,

$$\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot\frac{A}{2}$$

$$\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot\frac{B}{2}$$

$$\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot\frac{C}{2}$$

(vii) Deductions from Tangent Rule

$$\frac{b-c}{b+c} = \tan\left(\frac{B-C}{2}\right) \tan\frac{A}{2}$$

$$\frac{a-b}{a+b} = \tan\left(\frac{A-B}{2}\right) \tan\frac{C}{2}$$

$$\frac{c-a}{c+a} = \tan\left(\frac{C-A}{2}\right) \tan\left(\frac{B}{2}\right)$$

+ If a, b, c are the sides of Δ and semiperimeter is $s = \dfrac{a+b+c}{2}$, then area of a triangle is given by,

$$\Delta = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B = \frac{1}{2} ab \sin C$$

$$\Delta = 2R^2 \sin A \sin B \sin C$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Delta = \frac{abc}{4R}$$

+ **Half Angle Formulae of Triangle**

(i) $\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\sin\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$,

$\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(ii) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{bc}}$,

$\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{bc}}$

(iii) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$, $\tan\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$,

$\tan\frac{C}{2} = \sqrt{\frac{(s-b)(s-a)}{s(s-c)}}$

Deduction from $\frac{1}{2}$ Angle Formulae

+ $\sin A = 2\sin\frac{A}{2}\cos\frac{A}{2} = \frac{2\Delta}{bc}$
+ $\sin B = 2\sin\frac{B}{2}\cos\frac{B}{2} = \frac{2\Delta}{ca}$
+ $\sin C = 2\sin\frac{C}{2}\cos\frac{C}{2} = \frac{2\Delta}{ab}$
+ $\tan\left(\frac{A}{2}\right) = \frac{(s-b)(s-c)}{\Delta} = \frac{\Delta}{s(s-a)} = \frac{1}{\cot\left(\frac{A}{2}\right)}$
+ $\tan\left(\frac{B}{2}\right) = \frac{(s-a)(s-c)}{\Delta} = \frac{\Delta}{s(s-b)} = \frac{1}{\cot\left(\frac{B}{2}\right)}$
+ $\tan\left(\frac{C}{2}\right) = \frac{(s-a)(s-b)}{\Delta} = \frac{\Delta}{s(s-c)} = \frac{1}{\cot\left(\frac{C}{2}\right)}$

From Sine Rule in a Triangle,

+ $\frac{a+b}{c} = \frac{\cos\left(\frac{A-B}{2}\right)}{\sin\left(\frac{C}{2}\right)}, \quad \frac{a-b}{c} = \frac{\sin\left(\frac{A-B}{2}\right)}{\cos\left(\frac{C}{2}\right)}$
+ $\frac{b+c}{a} = \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\left(\frac{A}{2}\right)}, \quad \frac{b-c}{a} = \frac{\sin\left(\frac{B-C}{2}\right)}{\cos\left(\frac{A}{2}\right)}$
+ $\frac{c+a}{b} = \frac{\cos\left(\frac{C-A}{2}\right)}{\sin\left(\frac{B}{2}\right)}, \quad \frac{c-a}{b} = \frac{\sin\left(\frac{C-A}{2}\right)}{\cos\left(\frac{B}{2}\right)}$
+ If a, b, c are in A.P, then $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in A.P.
+ If a, b, c are in H.P, then $\sin^2\frac{A}{2}, \sin^2\frac{B}{2}, \sin^2\frac{C}{2}$ are in H.P.
+ In a Δ if $\angle A = 60°$, then
  (i) $\frac{1}{a+b} + \frac{1}{a+c} = \frac{3}{a+b+c}$
  (ii) $(a+b+c)(b+c-a) = 3bc$
  (iii) $\frac{c}{a+b} + \frac{b}{a+c} = 1$

  The respective results follow for $\angle B = 60°$ and $\angle C = 60°$.

Incircle and Excircle

+ The circle which touches all three sides of a triangle internally is referred as incircle or inscribed.
+ The centre (I) and radius (r) of an incircle is referred as in centre and inradius respectively.

In a triangle ABC,

+ $\tan\frac{A}{2}\tan\frac{B}{2} = \frac{s-c}{s}$

  $\tan\frac{C}{2}\tan\frac{A}{2} = \frac{s-b}{s}$

  $\tan\frac{B}{2}\tan\frac{C}{2} = \frac{s-a}{s}$

(i) $\Delta = rs$

(ii) $r = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (\text{s}$ ...

(iii) $r = \frac{a}{\cot\frac{B}{2} + \cot\frac{C}{2}} = \frac{b}{\cot\frac{C}{2} + \cot\frac{A}{2}} = \frac{c}{\cot\frac{A}{2} + \ldots}$

(iv) $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

+ The circle that touches the sides AB ... externally and the side BC internally is ... as excircle opposite to angle A.
+ The radius and centre of circle opp... angle A are called exradius and ex... denoted by r_1 and I_1 respectively.

  Similarly r_2, r_3 and I_2, I_3 are defined.

  (i) $r_1 = \frac{\Delta}{s-a}, \; r_2 = \frac{\Delta}{s-b}, \; r_3 = \frac{\Delta}{s-c}$

  (ii) $r_1 = s\tan\frac{A}{2} = (s-c)\cot\frac{B}{2} = (s-b)\ldots$

  (iii) $r_1 = \frac{a}{\tan\frac{B}{2} + \tan\frac{C}{2}}$

  $r_2 = \frac{b}{\tan\frac{C}{2} + \tan\frac{A}{2}}$

  $r_3 = \frac{c}{\tan\frac{A}{2} + \tan\frac{B}{2}}$

  (iv) $r_1 = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

  $r_2 = 4R\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$

  $r_3 = 4R\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

+ If in a triangle r_1, r_2, r_3 are the exradii the...

  $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$

  $r r_1 r_2 r_3 = \Delta^2$

  $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2+b^2+c^2}{\Delta^2}$

+ If a, b, c are in A.P, if and only if r_1, r_2, r_3 ... H.P.
+ $r + r_3 + r_2 - r_1 = 4R\cos A$
+ $r + r_3 + r_1 - r_2 = 4R\cos B$
+ $r + r_1 + r_2 - r_3 = 4R\cos C$
+ If p_1, p_2, p_3 are altitudes drawn from the ver... of A, B, C to opposite sides of a triangle...

  (i) $\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{1}{r}$

  (ii) $\frac{1}{p_2} + \frac{1}{p_3} - \frac{1}{p_1} = \frac{1}{r_1}$

  (iii) $p_1 p_2 p_3 = \frac{(abc)^2}{8R^3} = \frac{8\Delta^3}{abc}$

  (iv) $\frac{1}{p_1^2} + \frac{1}{p_2^2} + \frac{1}{p_3^2} = \frac{\cot A + \cot B + \cot C}{\Delta}$